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Combined paleomagnetic and geochronological study on Cretaceous strata of the Qiangtang terrane, central Tibet

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ABSTRACT

A combined paleomagnetic and geochronological investigation has been performed on Cretaceous rocks in southern Qiangtang terrane (32.5°N, 84.3°E), near Gerze, central Tibetan Plateau. A total of 14 sites of volcanic rocks and 22 sites of red beds have been sampled. Our new U–Pb geochronologic study of zircons dates the volcanic rocks at 103.8 ± 0.46 Ma (Early Cretaceous) while the red beds belong to the Late Cretaceous. Rock magnetic experiments suggest that magnetite and hematite are the main magnetic carriers. After removing a low temperature component of viscous magnetic remanence, stable characteristic remanent magnetization (ChRM) was isolated successfully from all the sites by stepwise thermal demagnetization. The tilt-corrected mean direction from the 14 lava sites is $D = 348.0°$, $I = 47.3°$, $k = 51.0$, $\alpha_{95} = 5.6°$, corresponding to a paleopole at 79.3°N, 339.8°E, $A_{95} = 5.7°$ and yielding a paleolatitude of 29.3° ± 5.7°N for the study area. The ChRM directions isolated from the volcanic rocks pass a fold test at 95% confidence, suggesting a primary origin. The volcanic data appear to have effectively averaged out secular variation as indicated by both geological evidence and results from analyzing the virtual geomagnetic pole (VGP) scatter. The mean inclination from the Late Cretaceous red beds, however, is 13.1° shallower than that of the ~100 Ma volcanic rocks. After performing an elongation/inclination analysis on 174 samples of the red beds, a mean inclination of 47.9° with 95% confidence limits between 41.9° and 54.3° is obtained, which is consistent with the mean inclination of the volcanic rocks. The site-mean direction of the Late Cretaceous red beds after tilt-correction and inclination shallowing correction is $D = 312.6°$, $I = 47.7°$, $k = 109.7$, $\alpha_{95} = 3.0°$, $N = 22$ sites, corresponding to a paleopole at 49.2°N, 1.9°E, $A_{95} = 3.2°$ (yielding a paleolatitude of 28.7° ± 3.2°N for the study area). The ChRM of the red beds also passes a fold test at 99% confidence, indicating a primary origin. Comparing the paleolatitude of the Qiangtang terrane with the stable Asia, there is no significant difference between our sampling location in the southern Qiangtang terrane and the stable Asia during ~100 Ma and Late Cretaceous. Our results together with the high quality data previously published suggest that an ~550 km N–S convergence between the Qiangtang and Lhasa terranes happened after ~100 Ma. Comparison of the mean directions with expected directions from the stable Asia indicates that the Gerze area had experienced a significant counterclockwise rotation after ~100 Ma, which is most likely caused by the India–Asia collision.

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1. Introduction

As the most extensive region of elevated topography on the Earth's surface, the Tibetan Plateau was formed by progressive accretion of several terranes to the stable Asian continent since Early Paleozoic (Allègre et al., 1984; Sengör, 1987; Dewey et al., 1988; Yin and Harrison, 2000; Metcalfe, 2006; Zhu et al., 2009, 2013; C.S. Wang et al., 2014; Q. Wang et al., 2014; Zhu et al., 2015). The tectonic processes of the Tibetan Plateau were also closely related to the opening, expansion, and closure of the Paleo-Tethyan and Neo-Tethyan oceans (Sengör et al., 1979; Sengör, 1987; Dewey et al., 1988; Sengör et al., 1988, 1993; Metcalfe, 1999) and climaxed by the Cenozoic collision between India and Asia that has forcefully raised the Tibetan Plateau (Molnar and Tapponnier, 1975; Aitchison et al., 2011; Dai et al., 2012; Guan et al., 2012; Xu et al., 2012), the most extensive region of elevated topography on the Earth's surface. Thus, the tectonic history of the plateau has attracted worldwide attention as it contains a complete record of subduction, collision and intra-continental convergence (Zhang et al., 2010; Cao et al., 2011; Xia et al., 2011; Hébert et al., 2012; Replumaz et al., 2013; K.J. Zhang et al., 2014; Z.M. Zhang et al., 2014). The Qiangtang terrane, one of the major terranes of the Tibetan Plateau, accreted to the Asia

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continent in the Late Triassic and collided with the Lhasa terrane during the Jurassic–Cretaceous, accompanied by the closure of the Bangong–Nujiang Tethyan Ocean (Kapp et al., 2000; Yin and Harrison, 2000; Kapp et al., 2003; Guynn et al., 2006; Zhu et al., 2006; Kapp et al., 2007; Wu et al., 2011; Zhang et al., 2012; Li et al., 2013; Zhu et al., 2013; Fan et al., 2014; Li et al., 2014; C.S. Wang et al., 2014; Q. Wang et al., 2014; K.J. Zhang et al., 2014; Z.M. Zhang et al., 2014; Fan et al., 2015; Li et al., 2015a,b; Zhu et al., 2015). It is generally accepted that the eastern and western segments of the Bangong–Nujiang suture zone experienced diachronous convergent processes (Dewey et al., 1988; Matte et al., 1996; Yin and Harrison, 2000; Wang et al., 2005; C.S. Wang et al., 2014; Q. Wang et al., 2014). The closure time for the western segment, however, remains controversial, ranging from Middle Jurassic (Kapp et al., 2003, 2007; H.F. Zhang et al., 2007; Y.X. Zhang et al., 2007; Qu et al., 2012; Li et al., 2014; K.J. Zhang et al., 2014; Z.M. Zhang et al., 2014) to Early or Middle Cretaceous (Wang et al., 2005; Zhu et al., 2006; Zhang et al., 2012; Li et al., 2013; Wang and Wei, 2013; Fan et al., 2014; C.S. Wang et al., 2014; Q. Wang et al., 2014; K.J. Zhang et al., 2014; Z.M. Zhang et al., 2014; Fan et al., 2015; Li et al., 2015a; Zhu et al., 2015). Continued north–south narrowing along the Bangong–Nujiang suture during the Cretaceous has been attributed to the substantial Cretaceous deformation in central Tibet (Yin and Harrison, 2000; Kapp et al., 2005; Wang et al., 2005; Li et al., 2013; Fan et al., 2014; C.S. Wang et al., 2014; Q. Wang et al., 2014; Fan et al., 2015; Li et al., 2015a). Likewise, the continental shortening in the central Tibet prior to the Indian–Asian collision is also hotly debated (Murphy et al., 1997; Yin and Harrison, 2000; Kapp et al., 2003; Guynn et al., 2006; Pan et al., 2006; Zhu et al., 2006; Kapp et al., 2007; Zhu et al., 2009, 2011). In addition, vertical-axis rotations have been shown to be a common consequence of crustal deformation in settings with thrusting motions and bounding faults in major Asian tectonic terranes (Zhao and Coe, 1987; Chen et al., 1991, 1993; Rumelhart et al., 1999; Cogné et al., 2013). Many geological and paleomagnetic data support that the eastern margin of the Qiangtang terrane had experienced a large scale clockwise rotation since Cretaceous (England and Molnar, 1990; Otofuji et al., 1990; Huang et al., 1992; Xu et al., 2006; Otofuji et al., 2007, 2010), while the rotation models of other parts are still puzzling. The paleogeographic position of the Qiangtang terrane in Cretaceous thus has very important significance in studying the evolution of the Bangong–Nujiang Ocean, the continental shortening and the rotation model of the central Tibet caused by the collision between the Qiangtang–Lhasa terranes.

Paleomagnetism remains the only method that can quantify the paleolatitudinal position and the amount of rotation of a plate. Previously published late Mesozoic and Cenozoic paleomagnetic studies have played a key role in understanding the tectonic evolution of the Tibetan Plateau. Nevertheless, there are only a few Cretaceous paleomagnetic investigations in the Qiangtang terrane (Otofuji et al., 1990; Huang et al., 1992; Chen et al., 1993). Many of these results have the problems of inaccurate age of the sampled rocks, small number of samples and bad statistic parameters; and besides, all available data were obtained from sedimentary rocks, which need test of possible influence of the paleomagnetic inclination shallowing. More investigations and high quality data are therefore in demand.

In this paper, we report the new results obtained from a combined study including the geochronology and the paleomagnetism on ~100 Ma volcanic rocks and Late Cretaceous red beds outcropping in an area close to Gerze, southern Qiangtang terrane. We use our new data together with a selection of the previously published paleomagnetic data to discuss the paleogeographic position of the Qiangtang terrane, local rotations and the tectonic history of the Bangong–Nujiang suture.

2. Geological setting and sampling

The sampling area (32.5°N, 84.3°E) is ~30 km northeast to the Gerze town, being located in the southern margin of the Qiangtang terrane and on the north side of the mid-west portion of the Bangong–Nujiang suture (Fig. 1a). The strata in this area mainly contain Triassic, Jurassic, Cretaceous and Cenozoic. The Triassic and Jurassic strata are largely deformed and unconformably overlain by the Cretaceous strata, which, in turn, are unconformably overlain by Cenozoic Kangtuo Formation. Our sampling focused on the Cretaceous strata (Fig. 1b), which mainly

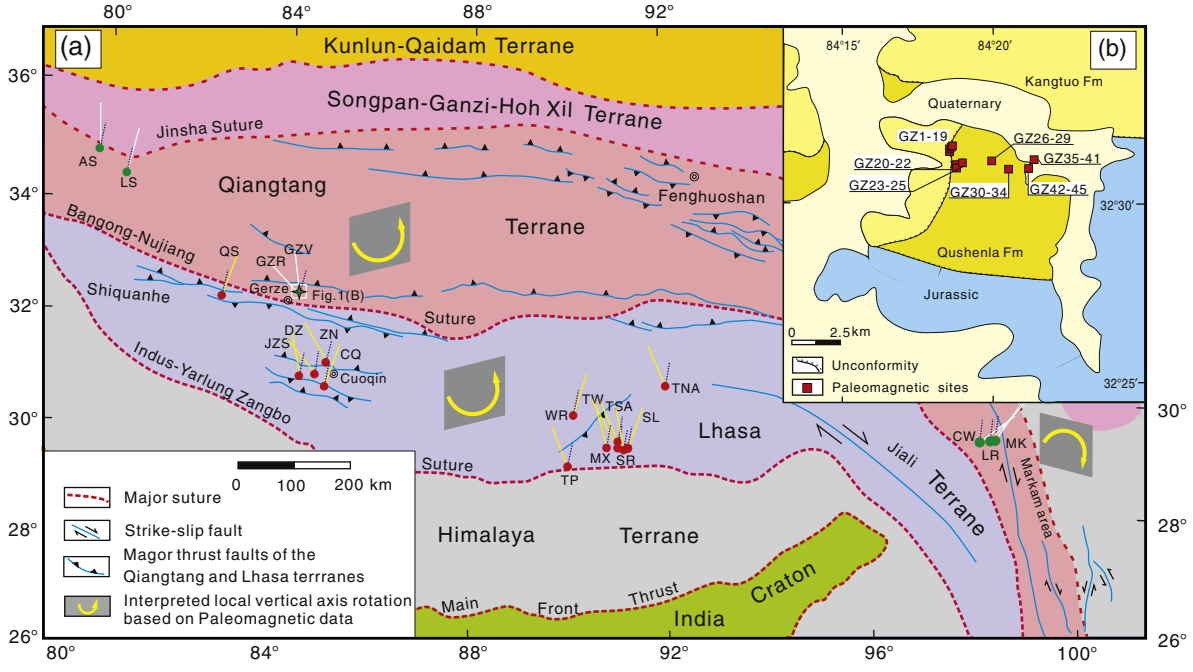


Fig. 1. (a) Simplified tectonic map of the Tibetan Plateau showing the major terranes and main sutures (modified from Yin and Harrison, 2000). The green and red closed circles show the location of reliable Cretaceous paleomagnetic investigations in the Qiangtang and Lhasa terranes, respectively (abbreviation showed in Table 2). Solid arrows represent the observed paleomagnetic declination, dotted lines represent the expected paleomagnetic declination calculated from the apparent polar wander path of Asia (reference site: 32.5°N, 84.3°E) (Cogné et al., 2013). (b) Simplified geological map of the sampling area and paleomagnetic sites (closed rectangle).

consist of volcanic rocks of andesite, basaltic andesite, basalt and rhyolite, with some red beds in the upper/top portion of this sequence (Fig. 2). Age and correlation of these strata had long been a matter of debate. It was mapped as Eocene strata in the 1:250,000 scale regional geological map of the Gerze area, but Chang et al. (2011) obtained the U–Pb zircon ages of ~110 Ma from the rhyolite rock of this succession in Rena-Co area (~20 km north to our sampled section) and these volcanic-clastic strata were correlated to Early Cretaceous Qushenla Formation in the 1:50,000 scale regional geological map of the Gerze area. The progress and deputes compel us to date the paleomagnetic samples in our own study. Our results indicate that the volcanic rocks are of Early Cretaceous, named Qushenla Formation following the 1:50,000 scale regional geological map. The Qushenla Formation used here is thus synchronous to the Meiriqiecuo Formation in the southern Qiantang terrane (Zhu et al., 2015) and the Zenong group in the Lhasa terrane (Chen et al., 2012; Ma et al., 2014). Details of our new geochronological analysis will be reported in the Section 3 of this paper.

Our sampled section contains four lithostratigraphic units (Fig. 2). The bottom unit, Member 1 of the Qushenla Formation, is mainly composed of basaltic andesite containing narrow phenocrysts. A total of 169 core samples from 15 sites (GZ30-45) were collected from the strata with SW dipping direction and dip of 26°–31°. The overlying unit (Member 2) consists of andesite and sandy conglomerate (Fig. 2) with NW dipping direction and dips of 15°. This unit contains two quenching edges clearly indicating boundaries for two lava subunits (Figs. 2 and 3a). A total of forty-five core samples were collected from the andesite of Member 2. The fact that the sandy conglomerate layer in Member 2 is sandwiched in the lava flows strongly argue that the emplacement of the volcanic units at the sampling locality spans sufficient time to likely average out the geomagnetic secular variation (Fig. 2). Member 3 is characterized by black basaltic andesites with well developed square-shaped phenocrysts. 34 cores samples were collected from this unit with dips of ~23° and SW dipping direction. In addition, several block samples from the fresh part of the basaltic andesite were collected for radiometric dating (Fig. 2). The uppermost unit, Member 4, comprises thin purple sandstone and siltstone, and are overlain with angular unconformably by the Paleogene Kangtuo Formation (40–30 Ma) (Zhong et al., 2008; Ding et al., 2015). A total of 242 core samples (22 sites) were collected from two sections of Member 4 (Fig. 2). 19 sites were collected from one section which the bedding dips steeply toward the NW at 77°–90° while the bedding of the other section dips at 18° toward the west. The age of the red beds has not been well constrained thus far. They can be any age between ~100 and ~40 Ma, although they are considered as part of the Qushenla Formation in the 1:50,000 scale regional geological map of the Gerze area.

Paleomagnetic samples were collected using a gasoline-powered drill and a magnetic compass mounted on an orienting device. We corrected for the effect of local magnetic anomalies on magnetic orientation by taking solar azimuths using a sun compass. For most samples, the discrepancy between the two compass readings is within 3°. All cores were cut into 2.2-cm-length specimens in the laboratory for further measurements.

3. Zircon geochronology

3.1. Analytical techniques

Geochronological samples of the fresh volcanic rocks were collected for LA-ICP-MS zircon U–Pb dating. Zircons were separated by heavy-liquid and magnetic methods. More than 1000 grains were hand-picked under a binocular microscope from the basaltic andesite samples. The representative grains together with a standard zircon were coined in an epoxy-resin mount and polished to expose the crystal interiors. Then the polished mounts were photographed under both transmitted and reflected light and cathodoluminescence (CL). Mount was vacuum-coated with a layer of gold. The U, Th and Pb isotope compositions were analyzed using laser ablation multicollector inductively coupled plasma mass spectrometry (LA-ICP-MS) at the Tianjin Institute of Geology and Mineral Resources, following the procedures described by Yuan et al. (2003).

3.2. Ages of the basaltic andesite

Zircons separated from the basaltic andesite sample are euhedral grains that are >200 μm in length (Fig. 4a), with a clearly identifiable regular banded structure. The shapes of the zircons are the characteristics of a column or a needle, strongly indicating a magmatic origin (Fig. 4a) (Li, 2009). A total of 40 zircon grains were analyzed, each with a probe spot.

The cumulative probability plot displays a population around 104 Ma (see supplementary data, Table A1; Fig. 4c). The results yielded a weighted-mean $^{206}Pb/^{238}U$ date of 103.8 ± 0.46 Ma with a mean square of weighted deviates (MSWD) of 0.90 (Fig. 4b, c, d). We thus consider that the best estimate age of the volcanic rocks is 103.8 ± 0.46 Ma. It is consistent with the results previously reported by Chang et al. (2011). We conclude that the ages of our sampled volcanic rocks range from ~110 to ~100 Ma, and the red beds are of Late Cretaceous.

4. Paleomagnetic laboratory techniques and measurements

All specimens were subject to stepwise thermal demagnetization with an ASC-TD48 furnace (with an internal residual field less than 10 nT). Remanent magnetization measurements were carried out using a 2G-755-4K cryogenic magnetometer for the red sandstones and an AGICO JR-6A spinner magnetometer for the strongly magnetized volcanic rocks, in the Paleomagnetism and Environmental Magnetism Laboratory (PMEML) of the China University of Geosciences, Beijing (CUGB). All the instruments are located in a shielded room with residual fields less than 200 nT. Demagnetization temperature intervals are generally large (50–30 °C) in the low temperature part, and become smaller (10–5 °C) at higher temperatures. In order to better understand magnetic mineralogy and help interpret the paleomagnetic data, hysteresis loops were performed on representative specimens. A KLY-3S Kappabridge susceptibility system was used to measure the anisotropy of magnetic susceptibility (AMS) to assess if the rocks have been affected by tectonic strain. The experiment acquiring the hysteresis loops was performed in the Institute of Geophysics, China Earthquake Administration, Beijing.

Characteristic remanent magnetization (ChRM) directions were determined for all the specimens using principal component analysis (Kirschvink, 1980). Site-mean paleomagnetic directions were calculated using Fisher statistics (Fisher, 1953) and paleomagnetic data were analyzed using R. Enkin's and J.P. Cogné's program packages (Enkin, 1990; Cogné, 2003).

5. Rockmagnetic and paleomagnetic results

5.1. Volcanic rocks

Anisotropy of magnetic susceptibility (AMS) was firstly measured using a KLY-3S Kappabridge to detect possible disturbances of the rock fabric. The AMS of 107 lava specimens show that the directions of the maximum, intermediate and minimum susceptibility axes are scattered without any regulation which means that the Cretaceous lava has not been disturbed by any tectonic deformation (Fig. 5a) (Hrouda, 1982).

Thermal demagnetization behavior of the volcanic rocks shows that the unblocking temperatures of majority of the specimens are close to 675 °C suggesting that hematite is the main magnetic carrier (Fig. 6a, b, d, e, f). Obvious decay of the remanent magnetization near 575 °C in these samples suggests the existence of the magnetite as well. For the remaining lava specimens, magnetite is the main magnetic mineral as

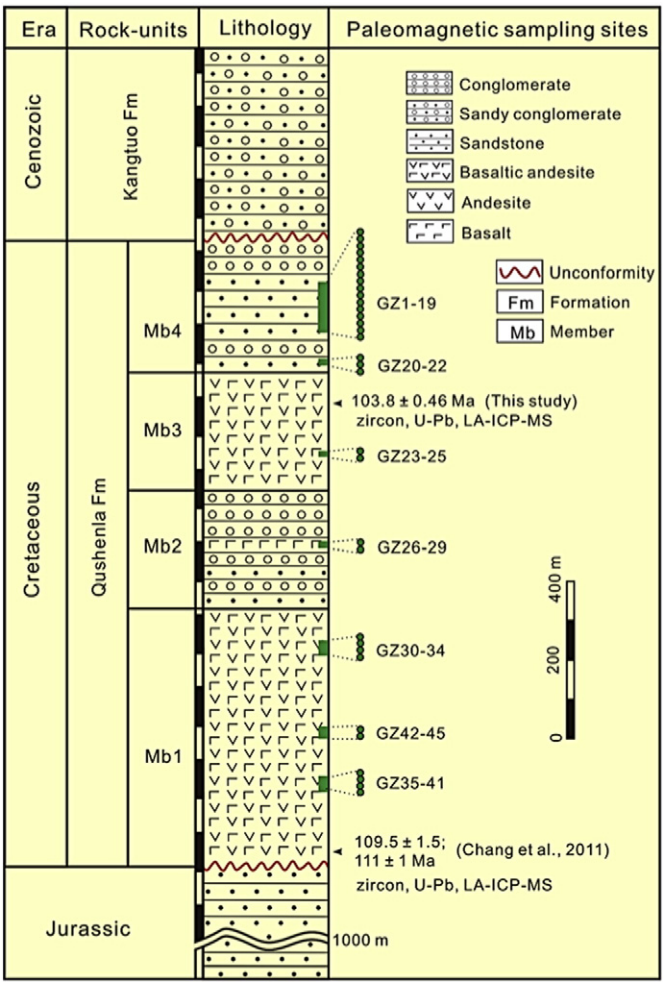


Fig. 2. Stratigraphic columns with paleomagnetic sampling sites, Gerze area.

indicated by unblocking temperature up to 575 °C (Fig. 6c). The magnetic minerals are supported by the hysteresis results from the represent specimens. Fig. 5c shows an obvious wasp-waisted shape with a saturation field higher than 800 mT, suggesting the existence of both high-coercivity and low-coercivity magnetic minerals (hematite and magnetite). For the magnetite dominate samples, the wasp-waisted characteristics are not present in the sample and the saturation field is generally lower than 400 mT (Fig. 5d), indicating the presence of low coercivity magnetic carriers. Combined with the thermal demagnetization behavior of the volcanic rocks, the fact that both magnetite and hematite are present in the volcanic rocks suggests that the hematite was probably formed by auto-oxidation processes during primary cooling of the volcanic rocks.

In majority of the lava specimens, a single component appears in the behavior of the stepwise thermal demagnetization that decayed toward the origin after removing a small and unstable viscous magnetization component at ~100 °C (Fig. 6a–d). However, some specimens of sites 25, 27, 33, and 38 show two components of magnetization (Fig. 6e, f). The low temperature component (LTC) was isolated by the 230 °C or 500 °C. The *in situ* directions generally cluster near the present geomagnetic field (PGF), indicating a recent viscous remanent magnetization (VRM) origin (Fig. 8a). The high temperature component (HTC) was resolved above 300 °C–500 °C and unblocked between 660 °C and 700 °C. The ChRM from the lava specimens shows a northwesterly direction with a moderate to steep-down inclination (Fig. 8c, d). The site-mean directions and the virtual geomagnetic poles (VGPs) corresponding to each site-mean direction are listed in Table 1. It is worth mentioning that some previous studies failed to identify bedding for the volcanic rocks. In this study, the attitudes of the lava were marked clearly by the quenching edges, air bubbles and phenocryst (Fig. 3a). Furthermore, the attitude of stratum along the quenching edges is consistent with the regional occurrence and could be used to make exactly structural correction for the volcanic rocks (Fig. 3; Table 2). The site-mean directions for 14 sites is $Dg = 11.4°$, $Ig = 39.9°$, $kg = 39.2$, $\alpha_{95} = 6.4°$ *in situ*, and $Ds = 348.0°$, $Is = 47.3°$, $ks = 51.0$, $\alpha_{95} = 5.6°$ after bedding correction (Fig. 8e, f; Table 2).

The paleomagnetic directions from the volcanic rocks are the instantaneous records of the geomagnetic field and free of inclination flattening effects. However, the results sometimes may be unable to average out the paleosecular variation (PSV) of the geomagnetic field. To test the PSV, a widely used approach is to calculate the statistical scatter of the VGPs under the condition that each VGP is a spot record of the geomagnetic field (e.g. Ma et al., 2014; Ren et al., 2016). On the premise of enough lava flows engaged in the final statistic, we obtain a VGP scatter of 10.6° which is consistent with the expected value (11.4°) with the 95% confidence level (9.0°, 14.0°) at 29.3°N during the Cretaceous Normal Superchron (CNS: 84–125 Ma) (Cox, 1970; McFadden et al., 1988; Biggin et al., 2008). Furthermore, the VGP scatter (11.5°) is also within the limit of VGP scatter (10.1°, 15.4°) according to the formula and standard proposed by McFadden et al. (1991). We thus consider that the average paleomagnetic results from the lavas (14 independent sites) are free of PSV influence.

The ChRM directions obtained from the volcanic rocks also pass the fold test of McFadden and McElhinny (1990) at 95% confidence: "Xi1" value in geographic coordinate is 5.9, and in stratigraphic coordinates 0.8, with the critical Xi 4.4 at 95% confidence level. Thus, the HTC is

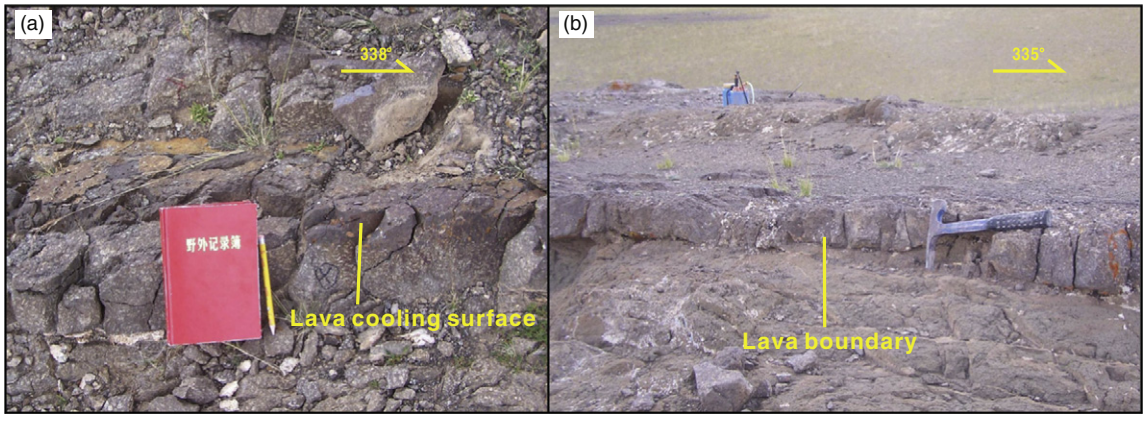


Fig. 3. (a) Quenching edge was found in the lavas, Member 1, Qushenla Formation. (b) Lava boundary.

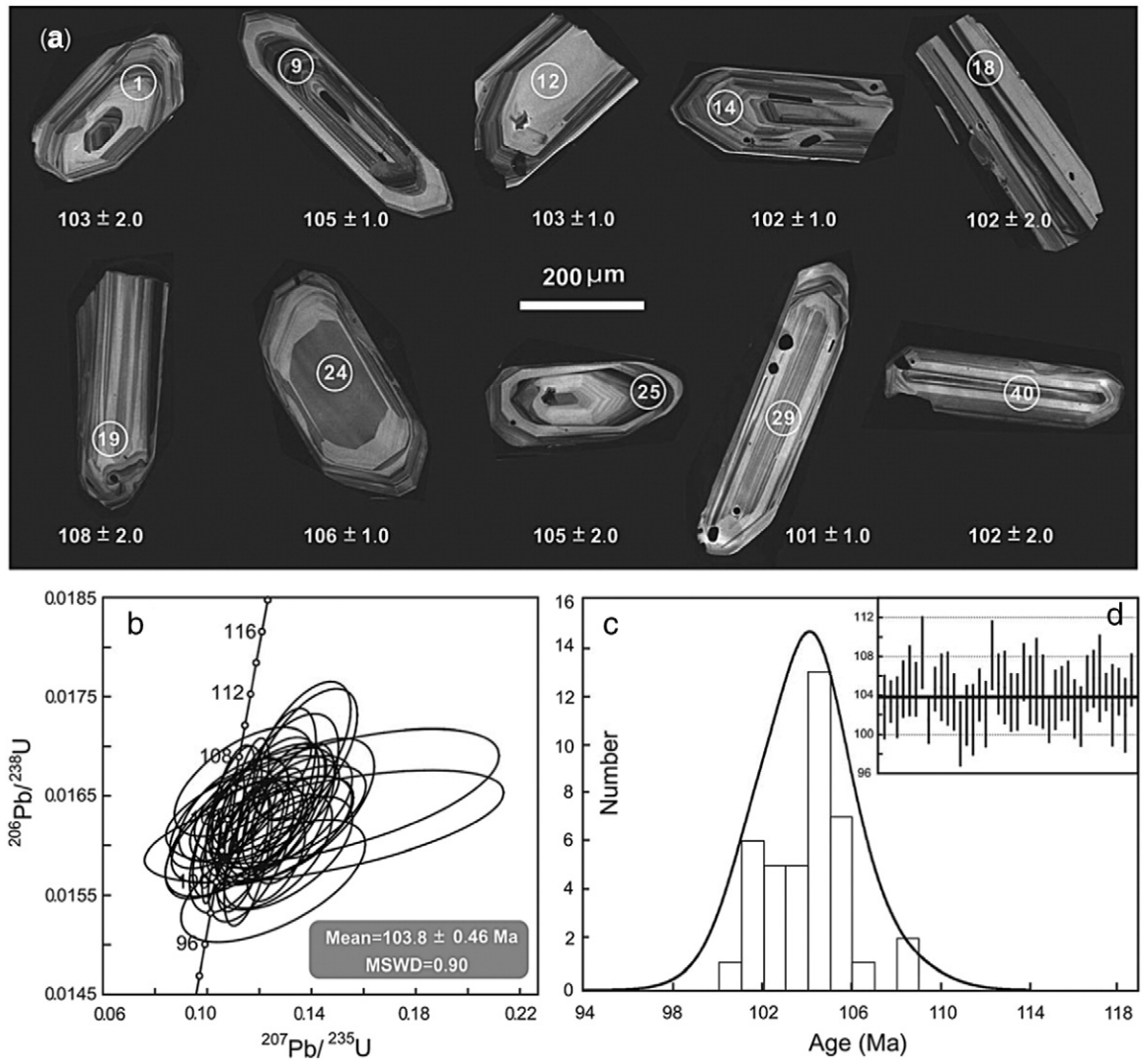


Fig. 4. (a) Cathodoluminescence (CL) images of representative zircons from the basaltic andesite of Member 3, Qushenla Formation, Gerze; circles show the position of the LA-ICP-MS spot analyses and the size of beam. (b) U–Pb concordia diagram of zircons from the basaltic andesite. (c) Cumulative probability plot showing the distribution of $^{206}Pb/^{238}U$ age, with a peak around 104 Ma. (d) Bar plot shows the weighted mean $^{206}U/^{238}Pb$ ages and vertical bar means the result of single zircon.

interpreted as a primary magnetization. Furthermore, the fact that all the inclinations isolated from the volcanic rocks are positive is consistent with magnetization within the Cretaceous Long Normal Superchron with an age between ~83.6 and ~126 Ma (Gradstein et al., 2012). These observations support that the ChRM directions are carried by primary thermal remanence that are reliable records of the ancient field during the cooling. Its corresponding paleopole is at 79.3°N, 339.8°E (A_{95} = 5.7°), yielding a paleolatitude of 29.3° ± 5.7°N for the study area (reference point: 32.5°N, 84.3°E) during ~110–100 Ma.

5.2. Red beds

The AMS of the red beds (Fig. 5b) in the stratigraphic coordinates show that the most minimum susceptibility axes (K_3) are perpendicular to the bedding plane, while maximum (K_1) and intermediate axes (K_2) are dispersed on a bedding plane. The results indicate that the red beds preserve a normal sedimentary magnetic fabric and the rocks have not experienced tectonic deformation since deposition (Tarling and Hrouda, 1993).

Most specimens of the red beds display similar unblocking-temperature near 660 or 680 °C with a sharp decay of the remanent magnetization around 575 °C (Fig. 7), indicating that these specimens may be dominated by hematite and magnetite. Furthermore, the hysteresis loops of the red beds show a wasp-waisted shape and do not close until a higher field approximately 460 mT (Fig. 5e, f), further suggesting the existence of both high coercivity and low coercivity magnetic minerals. Therefore, both magnetite and hematite are main magnetic carriers in the red sediments.

174 out of 247 demagnetized specimens revealed a concordant high temperature ChRM after removal of the LTCs that were isolated between 70 °C and 300 °C. For some of these specimens, the LTC is accounted for 30% intensity of the natural remanence (NRM) and clustered well around the PGF *in situ* for most specimens (Figs. 7 and 8b). But for other specimens, a soft magnetic component was removed by 100 °C and their directions are far from the PGF. We interpret that the LTCs of the red beds are probably a mixture of the recent field VRM and a random weak VRM obtained during preparing the specimens (Fig. 8b).

Directions of the HTC were obtained between 550 and 680 °C, pointing northwest and down after tilt-correction (Figs. 7 and 8h, j; Table 1). The site mean direction is Dg = 170.0°, Ig = 60.4°, kg = 8.0, α_{95} = 11.7°, N = 22, *in situ* and Ds = 312.6°, Is = 34.2°, ks = 98.8, α_{95} = 3.1°, N = 22, after bedding correction (Fig. 8i, j; Table 1).

Fig. 9a shows that the ChRM of the red beds exhibits an E–W elongated distribution evidently, which is an important sign of the

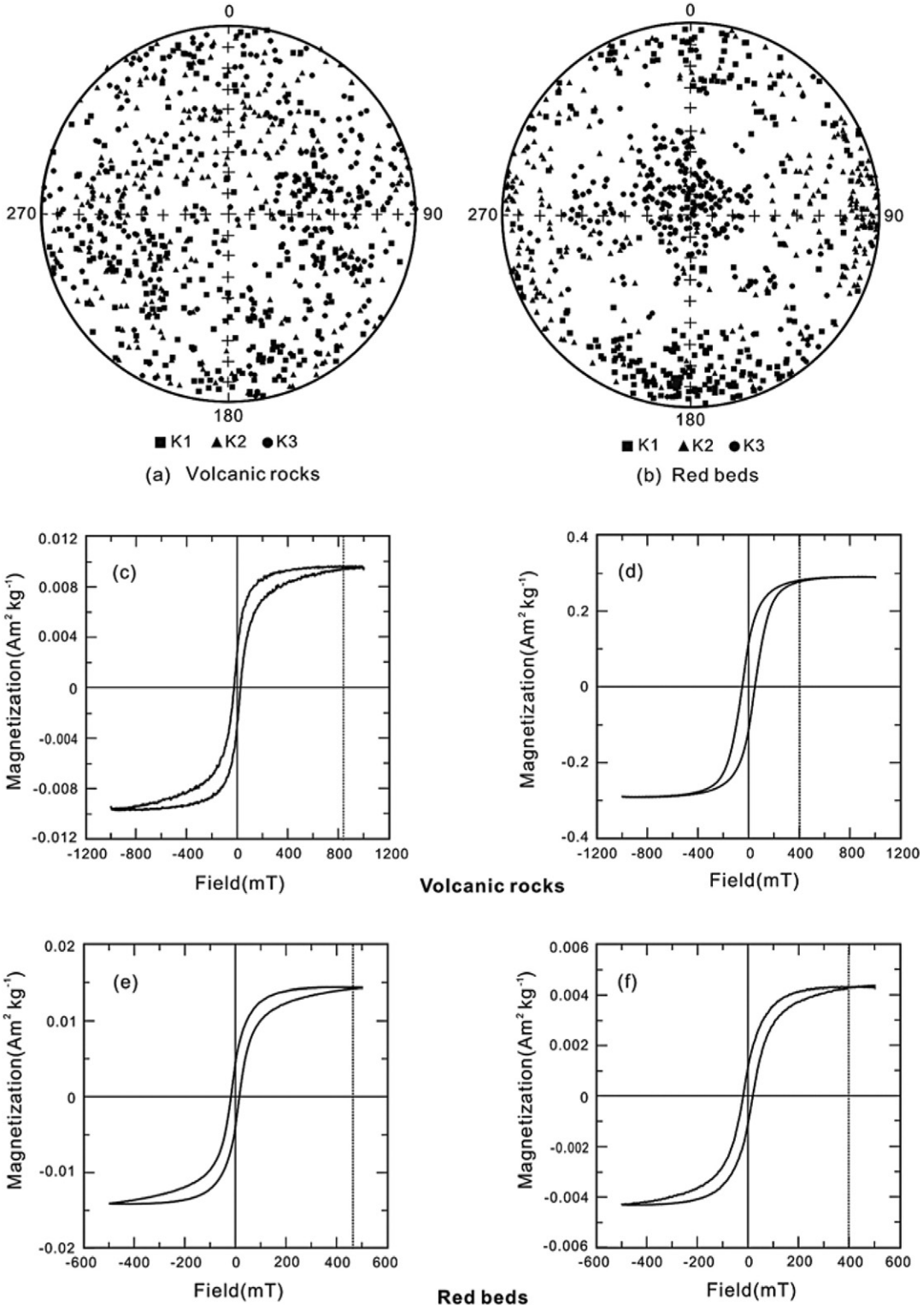


Fig. 5. Rock magnetic results of the representative samples from volcanic rocks and red beds, respectively. (a and b) Stereoplots (equal area, lower-hemisphere projection) of AMS principal axes of the volcanic rocks and red beds: K1/K2/K3, directions of maximum/median/minimum susceptibility axes. (c and d) Hysteresis loops of volcanic rocks. (e and f) Hysteresis loops of red beds.

inclination shallowing (Tauxe and Kent, 2004). We applied the elongation/inclination (E/I) method to the 174 directions of the red beds which show an east–west distribution evidently (Fig. 9a). The flattening correction factor (f) was 0.6. The inclination was corrected from 34.2° to 47.9°, with 95% confidence interval between 41.9° and 54.3° (Fig. 9b–e). The site-mean direction after the E/I correction is $Ds = 312.6°$, $Is = 47.7°$, $ks = 109.7$, $\alpha_{95} = 3.0°$, $N = 22$, in stratigraphic coordinates (Fig. 9c). These directions passed the fold test of McElhinny (1964) at 99% confidence level and the fold test of McFadden and McElhinny (1990) at 99% confidence level. The value of precise parameter "k" reaches maximum at 97.2% unfolding in the stepwise unfolding test (Fig. 9f; Watson and Enkin, 1993). These all suggest that the HTC can

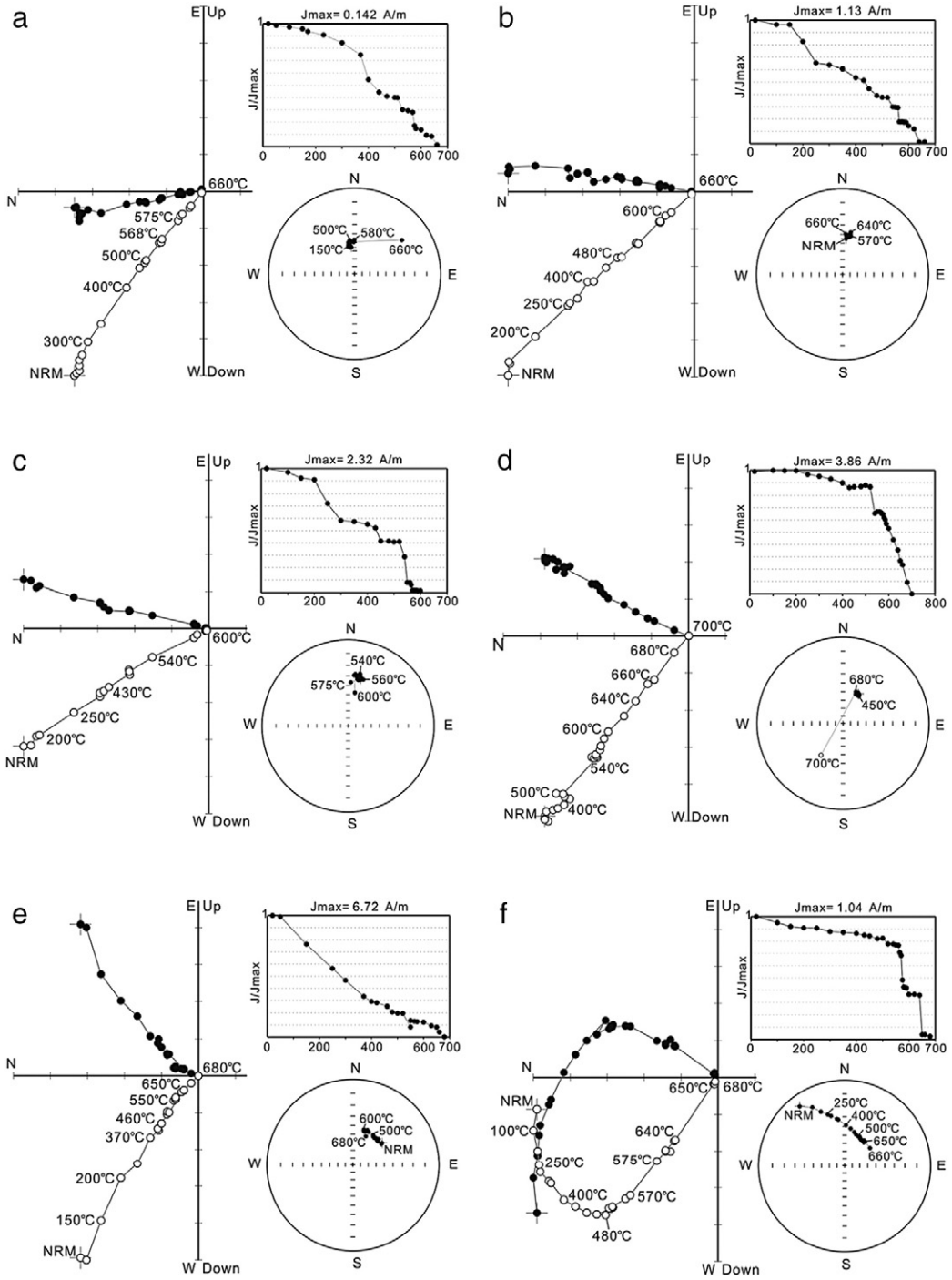


Fig. 6. Typical orthogonal vector plots in geographic coordinates for volcanic rocks. Solid (open) symbols represent the projections onto the horizontal (vertical) plane. In steroplots, closed (open) symbols represent positive (negative) inclination. NRM: natural remanent magnetization.

be interpreted as a pre-folding primary magnetization. The paleopole can be obtained by averaging all VGPs, at 49.2°N, 1.9°E, $A_{95} = 3.2°$, yielding the paleolatitude of 28.7° ± 3.2°N for the sampling area.

6. Tectonic implications

6.1. Cretaceous paleolatitude of the Qiangtang terrane

The available Cretaceous poles from the Qiangtang and Lhasa terranes are listed in Table 2 and plotted in Fig. 10a with the sampling location shown in Fig. 1a. There are five previously published poles, with the sampled areas in the southeast (Otofuji et al., 1990; Huang et al., 1992) and northwest of the Qiangtang terrane (Chen et al., 1993). The Cretaceous paleomagnetic data overall are still scarce in the central Qiangtang terrane and significant discrepancy between the different studies should be taken into account. The paleolatitude observed from the ~110–100 Ma lava in this study is well consistent with the paleolatitudes inferred from the three Early Cretaceous paleomagnetic poles of the southeastern Qiangtang terrane (poles CW, LR, MK in Table 2; Otofuji et al., 1990; Huang et al., 1992). But difference in the declinations between the two studies is significant, which might have resulted from some local rotations that will be discussed in Section 6.3. Noticeably, the inclinations observed from the Cretaceous red sandstones in the northwestern Qiangtang terrane (poles AS, LS in

Table 1
Site mean values and statistical parameters for the Cretaceous red beds and lava in Gaize area.

Site	*S/D*	*n/N*	Coor.	Direction				Palaeopole			
				D/(°)	*I*/(°)	*k*	α_{95}/(°)	Plat/(°N)	Plon/(°E)	dp/(°)	dm/(°)
Red Beds											
GZ1	235/86	10/11	G	162.5	55.8	90.0	5.1	−19.2	99.2	5.2	7.3
			S	312.3	36.2	90.0	5.1	45.9	350.2	3.5	5.9
			S*	312.4	50.3	97.9	4.9	49.9	5.5	4.4	6.6
GZ2	238/84	7/9	G	166.4	54.3	236.1	3.9	−21.5	96.3	3.9	5.5
			S	313.6	39.4	235.8	3.9	47.9	352.4	2.8	4.7
			S*	313.7	53.6	281.9	3.6	51.6	9.8	3.5	5.0
GZ3	234/95	5/9	G	160.9	63.2	62.6	9.7	−10.9	98.0	12.1	15.3
			S	315.4	20.6	62.5	9.8	43.6	336.6	5.4	10.3
			S*	315.4	31.6	48.6	11.1	47.0	344.4	7.0	12.5
GZ4	238/90	6/6	G	171.3	48.5	59.4	8.8	−21.9	81.5	3.8	5.4
			S	308.1	37.3	59.6	8.7	37.7	348.1	2.3	4.2
			S*	308.1	51.0	57.5	8.9	41.6	358.3	3.0	4.9
GZ5	238/90	6/6	G	181.4	52.8	131.5	5.9	−27.5	92.8	7.6	11.6
			S	304.8	30.1	131.6	5.9	37.7	350.0	3.6	6.6
			S*	304.9	43.7	125.9	6.0	41.8	0.9	4.7	7.5
GZ6	238/90	7/11	G	169.6	47.9	39.7	9.7	−27.8	94.6	8.3	12.7
			S	309.0	38.5	39.7	9.7	43.8	354.1	6.8	11.5
			S*	308.3	52.6	55.9	8.1	47.0	9.9	7.7	11.2
GZ7	235/86	9/12	G	171.4	55.5	54.4	7.0	−21.0	91.7	7.1	10.0
			S	306.6	34.2	54.6	7.0	40.5	351.9	4.6	8.0
			S*	306.3	48.1	68.9	6.2	44.2	5.0	5.3	8.1
GZ8	240/88	12/13	G	179.0	51.9	73.5	5.1	−25.0	85.2	4.8	7.0
			S	308.1	34.4	73.3	5.1	41.8	351.2	3.4	5.8
			S*	308.4	48.3	77.7	5.0	46.0	4.4	4.3	6.6
GZ9	236/89	13/17	G	160.1	51.3	69.9	5.0	−22.8	102.6	4.6	6.8
			S	314.2	38.3	69.6	5.0	48.1	351.0	3.5	5.9
			S*	314.4	52.4	92.7	4.3	52.0	7.8	4.1	5.9
GZ10	240/91	5/11	G	155.3	53.6	30.2	14.1	−19.4	105.8	13.7	19.7
			S	325.7	35.2	31.1	13.9	56.7	339.7	9.3	16.0
			S*	325.1	49.2	41.4	12.0	60.2	359.1	10.5	15.9
GZ11	234/90	8/13	G	166.8	52.1	95.6	5.7	−23.6	96.4	5.3	7.8
			S	306.6	34.4	96.1	5.7	40.5	352.0	3.7	6.5
			S*	306.7	48.4	109.9	5.3	44.7	5.2	4.6	7.0
GZ12	238/87	12/15	G	157.7	48.1	47.8	6.3	−24.8	105.7	5.4	8.2
			S	311.3	40.4	47.8	6.3	46.3	354.6	4.6	7.6
			S*	311.4	54.0	57.7	5.8	49.8	11.0	5.7	8.1
GZ13	240/91	13/20	G	160.5	53.4	89.2	4.4	−21.1	101.6	4.3	6.1
			S	318.0	38.5	91.2	4.4	51.3	348.8	3.1	5.2
			S*	318.1	52.7	125.9	3.7	55.0	7.3	3.5	5.1
GZ14	232/92	6/11	G	175.3	44.3	75.9	7.7	−31.3	89.2	6.1	9.7
			S	309.9	39.1	76.3	7.7	44.7	354.2	5.5	9.2
			S*	310.0	52.9	85.3	7.3	48.5	9.8	7.0	10.1
GZ15	240/90	8/13	G	154.1	62.7	29.5	10.4	−9.9	102.9	12.8	16.3
			S	327.3	27.2	29.5	10.4	55.2	330.8	6.2	11.3
			S*	326.6	38.9	22.5	11.9	58.6	343.2	8.4	14.2
GZ16	234/100	9/10	G	173.6	41.8	180.3	3.8	−33.1	91.3	2.8	4.7
			S	297.8	31.3	180.4	3.8	32.2	354.7	2.4	4.3
			S*	297.7	45.0	157.0	4.1	36.2	5.3	3.3	5.2
			S	308.1	26.5	69.1	8.1	39.4	345.5	4.8	8.8
			S*	308.1	39.1	59.6	8.8	43.2	355.1	6.3	10.5
GZ18	237/102	7/8	G	166.0	48.3	77.1	6.9	−26.8	98.0	5.9	9.0
			S	312.3	27.3	77.2	6.9	43.1	343.3	4.1	7.5
			S*	312.2	40.2	76.4	6.9	47.0	354.0	5.0	8.3
GZ19	237/102	7/8	G	168.2	43.7	94.2	6.3	−30.9	96.7	4.9	7.9
			S	308.5	30.9	93.6	6.3	41.1	348.3	3.9	7.0
			S*	308.4	44.4	86.5	6.5	45.0	0.1	5.1	8.2
GZ20	175/20	6/12	G	345.6	41.9	175.7	5.1	74.9	324.6	3.8	6.3
			S	329.9	36.0	176.0	5.1	60.4	336.8	3.4	5.9
			S*	329.9	50.2	178.4	5.0	64.4	359.0	4.5	6.7
GZ21	174/18	6/6	G	323.4	45.7	32.1	12.0	58.0	354.4	9.7	15.3
			S	311.0	34.8	32.2	12.0	44.3	349.8	7.9	13.8
			S*	311.5	47.9	26.9	13.2	48.5	2.7	11.2	17.2
GZ22	163/15	6/13	G	331.7	39.9	40.4	10.7	63.2	340.0	7.7	12.9
			S	320.4	35.6	40.3	10.7	52.5	344.3	7.2	12.4
			S*	319.9	49.3	45.9	10.0	55.9	1.3	8.8	13.3
Mean	$N = 22$		G	170.0	60.4	8.0	11.7	29.8	279.9	$A_{95} = 8.6°$	
	174/247		S	312.6	34.2	98.8	3.1	45.4	348.1	$A_{95} = 3.1°$	
			S*	312.6	47.7	109.7	3.0	49.2	1.9	$A_{95} = 3.2°$	
Lavas											
GZ23	164/23	6/6	G	9.7	41.9	418.2	3.3	78.1	216.2	2.5	4.0
			S	347.1	47.8	418.8	3.3	78.3	339.6	2.8	4.3

Table 1 (*continued*)

Site	*S/D*	*n/N*	Coor.	Direction				Palaeopole			
				D/(°)	*I*/(°)	*k*	α_{95}/(°)	Plat/(°N)	Plon/(°E)	dp/(°)	dm/(°)
GZ24–25	164/23	4/4	G	351.1	42.2	91.2	9.7	78.7	310.5	7.3	11.9
			S	330.4	40.8	91.1	9.7	62.4	342.5	7.1	11.8
GZ26–28	194/15	6/9	G	2.8	45.0	343.4	3.6	83.6	241.3	2.9	4.6
			S	347.6	45.3	346.5	3.6	77.8	329.6	2.9	4.6
GZ29	194/15	9/9	G	8.7	44.5	616.4	2.1	80.1	212.0	1.7	2.6
			S	353.5	46.3	622.7	2.1	82.6	315.0	1.7	2.7
GZ30	163/28	5/6	G	8.4	28.6	790.7	2.7	71.1	238.5	1.6	3.0
			S	350.7	36.8	792.5	2.7	75.4	301.3	1.8	3.2
GZ31	163/28	7/7	G	9.4	30.5	71.4	7.2	71.8	234.2	4.5	8.0
			S	350.4	39.0	71.1	7.2	76.5	305.8	5.1	8.6
GZ32*	166/28	4/4	G	16.2	23.7	112.0	8.7	65.0	224.2	4.9	9.3
			S	1.4	36.2	111.2	8.8	77.5	258.2	6.0	10.2
GZ33–34	166/28	7/8	G	12.8	30.5	145.5	5.0	70.2	225.5	3.1	5.6
			S	353.7	39.3	145.2	5.0	78.3	294.5	3.6	6.0
GZ35–36	158/31	9/12	G	27.5	49.1	267.5	3.2	66.4	173.0	2.8	4.2
			S	349.3	60.5	265.9	3.2	77.6	43.8	3.7	4.9
GZ37	158/31	8/10	G	30.1	50.3	836.1	1.9	64.5	169.4	1.7	2.5
			S	349.9	62.6	840.6	1.9	76.1	52.7	2.3	3.0
GZ38	158/31	8/12	G	30.8	47.6	207.5	3.9	63.3	174.3	3.3	5.1
			S	354.7	61.1	207.3	3.9	79.5	62.3	4.6	6.0
GZ39–41	158/31	9/9	G	27.9	45.1	323.6	2.9	65.1	180.6	2.3	3.7
			S	355.1	57.8	324.8	2.9	82.8	51.8	3.1	4.3
GZ42–43	152/26	5/5	G	0.5	38.6	195.9	5.5	79.2	261.8	3.9	6.5
			S	337.1	46.4	196.3	5.5	69.7	346.7	4.5	7.1
GZ44–45	152/26	4/4	G	353.7	32.2	204.8	6.4	73.9	286.5	4.1	7.2
			S	335.4	38.0	205.1	6.4	65.5	333.4	4.5	7.6
Mean	$N = 14$		G	11.4	39.9	39.2	6.4	76.0	213.2	$A_{95} = 6.6°$	
	91/105		S	348.0	47.3	51.0	5.6	79.3	339.8	$A_{95} = 5.7°$	

Notes: *n/N*: number of samples used to calculate mean/measured; *Dg/Ig*, declination/inclination in geographic coordinates; *Ds/Is*: declination/inclination in stratigraphic coordinates; D_{S^*}/I_{S^*}: declination/inclination after E/I correction; *k*: best estimate of the precision parameter of direction; α_{95}/A_{95}: radius of circle of 95% confidence on the direction/pole; GPlat/GPlon: latitude/longitude of VGP in the geographic coordinates; SPlat/SPlon: latitude/longitude of VGP in the stratigraphic coordinates; S*Plat/S*Plon: latitude/longitude of VGP after E/I correction; *S/D*: Strike/Dip; GZ32* (166, 28): the attitude of stratum of the quenching edges.
① Fold test for red beds is positive at the 99% confidence (McElhinny, 1964): $Ks/Kg = 13.7 > F(2^*(n2-1),(n1-1))$ at 1% point = 2.1.
② Fold test for red beds is also positive at the 99% confidence (McFadden and McElhinny, 1990): critical Xi at 99% = 7.7. Xi1 IS = 17.2, Xi1 TC = 2.6.
③ Fold test for volcanic rocks is positive at the 95% confidence (McFadden and McElhinny, 1990): critical Xi at 95% = 4.4. Xi1 IS = 5.9, Xi1 TC = 0.8.

Table 2) are obviously shallower than that of other studies (poles CW, LR, MK, GL and GR in Table 2) (Fig. 10a). We propose three hypotheses to explain this difference: (1) the poles AS and LS were influenced by inclination shallowing of sedimentary rock; (2) significant relative movement had once occurred between the NW end of the Qiangtang terrane and the middle-eastern portion of this terrane; or (3) the poles are different in age. They need further test.

The apparent polar wander paths for the stable Asia have been updated recently (e.g., Torsvik et al., 2012; Cogné et al., 2013). Many previous paleomagnetic studies of the eastern Asia had focused on the sediments that are often affected by possible inclination shallowing (Cogné et al., 1999; Gilder et al., 2001, 2003; Tan et al., 2003; Tauxe and Kent, 2004; Yan et al., 2005; Tan et al., 2007; Gilder et al., 2008; Dupont-Nivet et al., 2010a,b; Tan et al., 2010). To address this issue, Torsvik et al. (2012) applied a common correction factor value $f = 0.6$ to all the poles obtained from sedimentary rocks to establish their APWP. However, Ding et al. (2015) argued that the "*f*" value sometimes depends on the grain size of the red beds. Cogné et al. (2013) reanalyzed 533 Cretaceous and Cenozoic poles obtained from both sedimentary and volcanic rocks from the Asian blocks and suggested that there is an ~10° lower paleolatitude anomaly statistically observed from Late Cretaceous rocks distributing throughout the Asian continent (Fig. 10b). Although flattening of paleomagnetic inclination due to sedimentary processes in red beds is present, it seems to be not the only mechanism responsible for this and the Cenozoic inclination anomalies observed over the Asian continent (Wu et al., 2002). With this in mind, we adopt the new reference APWP constructed by Cogné et al. (2013) to discuss the relative motion between the Qiangtang terrane and other blocks in Asia.

As shown in Fig. 10b, the differences between our observed paleolatitudes and expected paleolatitudes from the data by Cogné et al. (2013), for the reference location (Gerze: 32.5°N, 84.3°E), are not significant, strongly suggesting that the Qiangtang terrane had become a part of the stable Asia by ~110 Ma.

6.2. Convergence between the Qiangtang and Lhasa terranes after ~100 Ma

Numerous studies suggested that the east and west segments of the Bangong–Nujiang suture experienced diachronous convergent processes, but the time of the closure of the Bangong–Nujiang Ocean has long been a subject of much debate (e.g., Dewey et al., 1988; Matte et al., 1996; Murphy et al., 1997; Yin and Harrison, 2000; Kapp et al., 2003; Guynn et al., 2006; Pan et al., 2006; Kapp et al., 2007; Zhu et al., 2009, 2011; Zhang et al., 2012; Zhu et al., 2013; Fan et al., 2014; C.S. Wang et al., 2014; Q. Wang et al., 2014; K.J. Zhang et al., 2014; Z.M. Zhang et al., 2014; Fan et al., 2015; Li et al., 2015a,b; Zhu et al., 2015). Comparing paleomagnetic results from coeval strata on both sides of the Bangong–Nujiang suture is critical to understanding the convergence history. Before that, some comments and a critical selection are necessary for the data from the Lhasa terrane.

The Cretaceous paleomagnetic data from volcanic and sedimentary rocks yield a wide range of paleolatitude of the Lhasa terrane, from 10.7° to 27.8°N (Table 2; Fig. 10b). The poles NL, QL and TL of Lin and Watts (1988) (Table 2) were defined by a small number of specimens, provided no field test and lack of precise determinations for structural attitude, thus they are excluded in further discussion. Pole SQ is derived from only 3 sites and has a large uncertainty. We thus ignore it too. We consider that the other poles of the Lhasa terrane listed in Table 2 are more reliable.

Poles TP, TW and TA are obtained from the Takena Formation red beds in the Linzhou basin and they are consistent with each other (Fig. 10a) (Pozzi et al., 1982; Westphal et al., 1983; Achache et al.,

Table 2
Summary of the Cretaceous paleomagnetic poles from the Qiangtang and Lhasa terranes.

Site location (°N/°E)	Mnemonic	Rock-units	Age (Ma)	Plat (°N)	Plon (°E)	A_{95} (°)	Palat (°N)	*N*/*n*	Criterion (*Q*)	Reference
Qiangtang terrane										
Gaize (32.5, 84.3)	GZR	Red beds	Late Cretaceous—103.8	49.2	1.9	3.2	28.7 ± 3.2	22/174	123 F5□7 (6)	This study
Gaize (32.5, 84.3)	GZV	Volcanic rocks	103.8 ± 0.46, U–Pb LA-ICP-MS	79.3	339.8	5.7	29.3 ± 5.7	14/91	123 F5□7 (6)	This study
Markham (29.7, 98.7)	MK	Mangkang Fm, red beds	Aptian–Turonian	56.7	172.7	9.6	27.5 ± 9.6	11/79	123 F5□7 (6)	Huang et al. (1992)
Markham (29.7, 98.4)	CW	Cuowa Fm, red beds	Berriasian–Barremian	40.6	170.5	11.9	23.1 ± 12.0	12/68	123F5D□ (6)	Huang et al. (1992)
Markham (29.7, 98.6)	LR	Laoran Fm, red sandstones	Barremian–Albian	48.5	175.8	9.5	22.8 ± 9.5	5/30	123□5□7 (5)	Otofuji et al. (1990)
Aksaichin (35.0, 79.7)	AS	Red sandstones	Albian–Aptian	66.3	256.5	5.1	9.0 ± 5.1	7/55	123F5D7 (7)	Chen et al. (1993)
Longmuco (34.5, 80.4)	LS	Red sandstones	Albian–Aptian	64.4	231.3	9.9	10.3 ± 9.9	4/50	123□5D7 (6)	Chen et al. (1993)
Lhasa terrane										
Cuoqin (31.2, 84.7)	JZS	Jingzhushan Fm, red beds	K2	49.0	344.3	5.3	18.0 ± 5.3	33/291	123F5D7 (7)	Yang et al. (2015)
Maxiang (29.9, 90.7)	MX	Shexing Fm, red beds Andesite	72.4–110, U–Pb SHRIMP	74.6	346.5	2.7	29.2 ± 2.7	100	123 F5□7 (6)	Sun et al. (2012), Lippert et al. (2014)
Qelico (31.7, 91.0)	*QL*	*Qelico lava*	*85–95, ^{40}Ar–^{39}Ar*	*74.0*	*318.0*	*14.6*	*22.3 ± 14.6*	*4/20*	*1□3□5□□ (3)*	Lin and Watts (1988)
Nagqu (31.5, 92.0)	*NL*	*Nagqu lava*	*95–100, ^{40}Ar–^{39}Ar*	*78.0*	*282.0*	*5.3*	*21.0 ± 5.3*	*9/33*	*123□□□□ (3)*	Lin and Watts (1988)
Linzhou (29.5, 91.1)	SR	Shexing Fm, red beds	Albian–Late Cretaceous	79.6	330.5	2.4	27.8 ± 2.4	43/196	123F5D□ (6)	Tan et al. (2010)
Linzhou (29.6, 91.2)	SL	Shexing Fm, lava	Albian–Late Cretaceous	69.1	191.7	4.2	24.3 ± 4.2	21/136	123F5D□ (6)	Tan et al. (2010)
Cuoqin (30.9, 84.9)	CQ	Andesite, tuff	93–99, ^{40}Ar–^{39}Ar	63.1	224.6	5.1	10.7 ± 5.1	10/87	123 F5□7 (6)	Tang et al. (2013)
Linzhou (29.2, 90.0)	TP	Takena Fm,red beds	Albian–Aptian	68.0	340.0	8.8	24.8 ± 8.8	7/68	123 F5□7 (6)	Pozzi et al. (1982)
Linzhou (29.9, 91.0)	TW	Takena Fm, red beds	Albian–Aptian	64.0	348.0	7.3	26.3 ± 7.3	6/57	123F5D7 (7)	Westphal et al. (1983)
North of Linzhou (31.0, 92.0)	TNA	Takena Fm, red beds	Albian–Aptian	63.5	325.4	6.5	17.4 ± 6.5	6/57	123 F5□7 (6)	Achache et al. (1984)
Linzhou (30.0 91.0)	TSA	Takena Fm, red beds	Albian–Aptian	71.2	288.4	7.9	15.1 ± 7.9	8/61	123 F5□□ (5)	Achache et al. (1984)
Mean TNA + TSA (rotated)	TA	Takena Fm, red beds	Albian–Aptian	63.4	323.2	4.8	16.6 ± 4.8	14/118	123 F5□7 (6)	Achache et al. (1984)
Mean for Takena Fm	TM	Takena Fm, red beds	Albian–Aptian	63.6	333.5	3.6	20.4 ± 3.6	27/243	123F5D7 (7)	Poles: TP, TW, TA
Linzhou (29.9, 91.2)	*TL*	*Takena Fm, red beds*	*Albian–Aptian*	*68.0*	*279.0*	*4.9*	*11.1 ± 4.9*	*8/51*	*123 F5□□ (5)*	Lin and Watts (1988)
Shiquanhe (32.7, 80.2)	*SQ*	*Limestones*	*Albian–Aptian*	*67.7*	*234.2*	*17.9*	*12.7 ± 17.9*	*3/62*	*123□5□7 (5)*	Chen et al. (1993)
Deqing (30.5, 90.1)	WR	Woronggou Gp, volcanics	114 ± 1.1, U–Pb SHRIMP	66.4	220.3	6.9	14.4 ± 6.9	16/88	123□5□7 (5)	Sun et al. (2008)
Cuoqin (31.1, 84.4)	DZ	Dianzhong Fm, volcanics	117–121, U–Pb LA-ICP-MS	70.5	292.9	7.4	15.0 ± 7.4	12/116	123□5D7 (6)	Yang et al. (2015)
Cuoqin (31.3, 85.0)	ZN	Zenong Gp, volcanic rocks	123.1 ± 0.9, U–Pb LA-ICP-MS	58.2	341.9	4.6	21.2 ± 4.6	18/162	123F5R7 (7)	Chen et al. (2012)
Yanhu (32.3, 82.6)	QS	Qushenla Fm, lava	120–132, U–Pb LA-ICP-MS	61.4	192.9	2.1	19.2 ± 2.1	51/444	123F5D7 (7)	Ma et al. (2014)

Notes: Site location: paleomagnetic sampling location; Mnemonic: pole name plotted in Figs. 1 and 9; Fm: Formation; Plat/Plon: latitude and longitude of a pole; A_{95}: radius of circle of 95% confidence on a pole; Palat: paleolatitude calculated using the reference point (32.5°N, 84.3°); *N*/*n*: site/sample number used for final statistic; Criterion (*Q*): data quality criteria and *Q* values (number of criteria met) according to the 7-point quality criterion system proposed by Van der Voo (1990) [1, well determined rock age; 2, sufficient number of samples, $n > 24$, k (or K) ≥ 10, $\alpha_{95} \leq 16.0°$; 3, step-wise demagnetization; 4 (F), positive fold test; 5, structural control and tectonic coherence with the craton discussed; 6, (R/D), positive reversals test/dual-polarity; 7, no resemblance to paleopole of younger age (by more than a period)]; "□": failed to meet this criterion. For poles GZR, MX, SR: the inclinations of the red beds have been corrected using the E/I method; TM: the mean paleomagnetic pole for the Takena red beds; italicized poles are excluded for tectonic interpretations (see text).

1984). We thus combine these three studies to calculate a mean paleomagnetic pole (TM, in Table 2; Fig. 10a) for the Takena red beds. This pole has an Albian–Aptian age and is located at 63.6°N, 333.5°E, $A_{95} = 3.6°$, yielding the paleolatitude of 20.4° ± 3.6°N for the reference point (Gerze, 32.5°N, 84.3°E). This paleolatitudinal result is in good agreement with those inferred from two high quality poles ZN (~123 Ma) and QS (~120–132 Ma). However, other two poles WR and DZ, both obtained from well-dated volcanic rocks, indicate obviously lower paleolatitudes for the Lhasa terrane (Table 2; Fig. 10b) (Sun et al., 2008; Yang et al., 2015).

For Late Cretaceous, there are three poles obtained from the Shexing Formation. Two of them (poles MX and SR, Table 2) are from red beds that occurred in the upper Shexing Formation. Lippert et al. (2014) has reviewed the pole MX and suggested an inclination shallowing correction for it. We adopt this correction and notice that the paleolatitudes inferred from these two poles are fairly close to that observed from the Gerze red beds by this study, being around 29°N. However, the other three poles (SL, JZS, CQ in Table 2) that are slightly or much older than the upper Shexing red beds yield obviously shallower paleolatitudes for the Lhasa terrane (Fig. 10b).

Our analysis reveals some paleolatitudinal variations for the Lhasa terrane in Cretaceous, although many need further test. There is no concisely coeval pole from the Lhasa terrane which can be compared to the Gerze lava pole (GZV in Table 2) of the Qiangtang terrane. The nearest ones

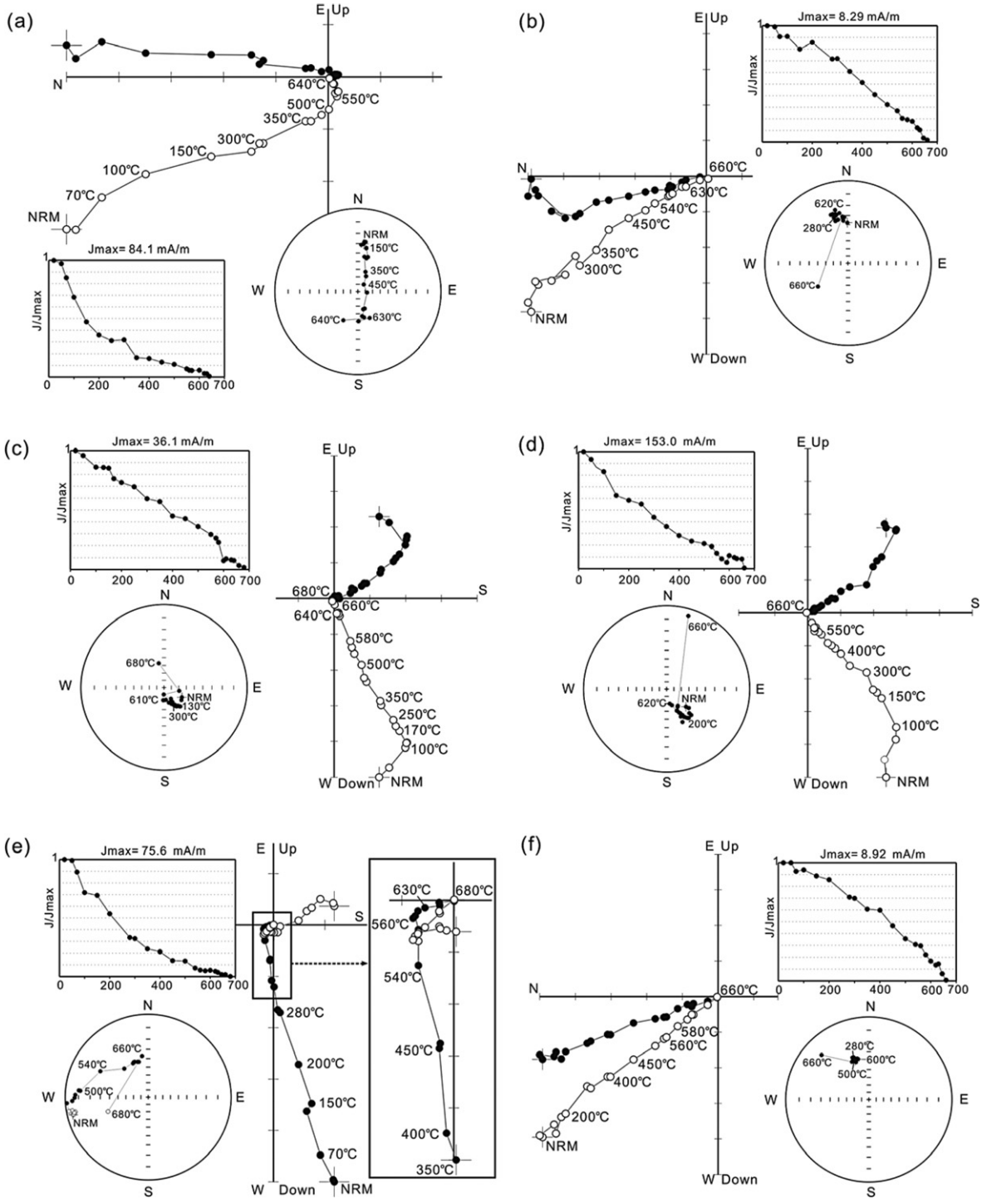


Fig. 7. Typical orthogonal vector plots in geographic coordinates for red beds. Solid (open) circles of the orthogonal vector plots represent the projections onto the horizontal (vertical) plane. In steroplots, closed (open) symbols represent positive (negative) inclination. NRM: natural remanent magnetization.

are Early Cretaceous pole TM and Late Cretaceous pole CQ (Table 2). The pole TM yields a paleolatitude of 20.4° ± 3.6°N for the reference location, ~10° lower than our observation. Meanwhile the pole CQ yields a paleolatitude of 10.7° ± 5.1°N for the reference point, almost ~20° lower than our observation. Our comparison indicates that there was considerable paleolatitudinal discrepancy between the Qiangtang terrane and the Lhasa terrane during the time interval around ~100 Ma. But a precisely quantitative discrepancy is difficult to be concluded at this stage, especially under the consideration of the complexities of the paleomagnetic data. Recently, Lippert et al. (2014) made a careful review over the data from Lhasa terrane and suggested that the Lhasa terrane had remained at ~24°N (recalculated to Gerze reference point) between ~110 and at least 50 Ma. If we adopted their analyses, the paleolatitudinal discrepancy between the Qiangtang and Lhasa terranes would be ~5°, during the time from ~110 Ma through Late cretaceous.

In summary, our new results suggest that the Gerze reference point (32.5°N, 84.3°E) of the southern Qiangtang terrane was located at 29.3° ± 5.7°N during ~110–100 Ma. It is at least ~5° higher than that inferred from the paleomagnetic results of the Lhasa terrane, suggesting that ~550 km N–S shortening between the Qiangtang and Lhasa

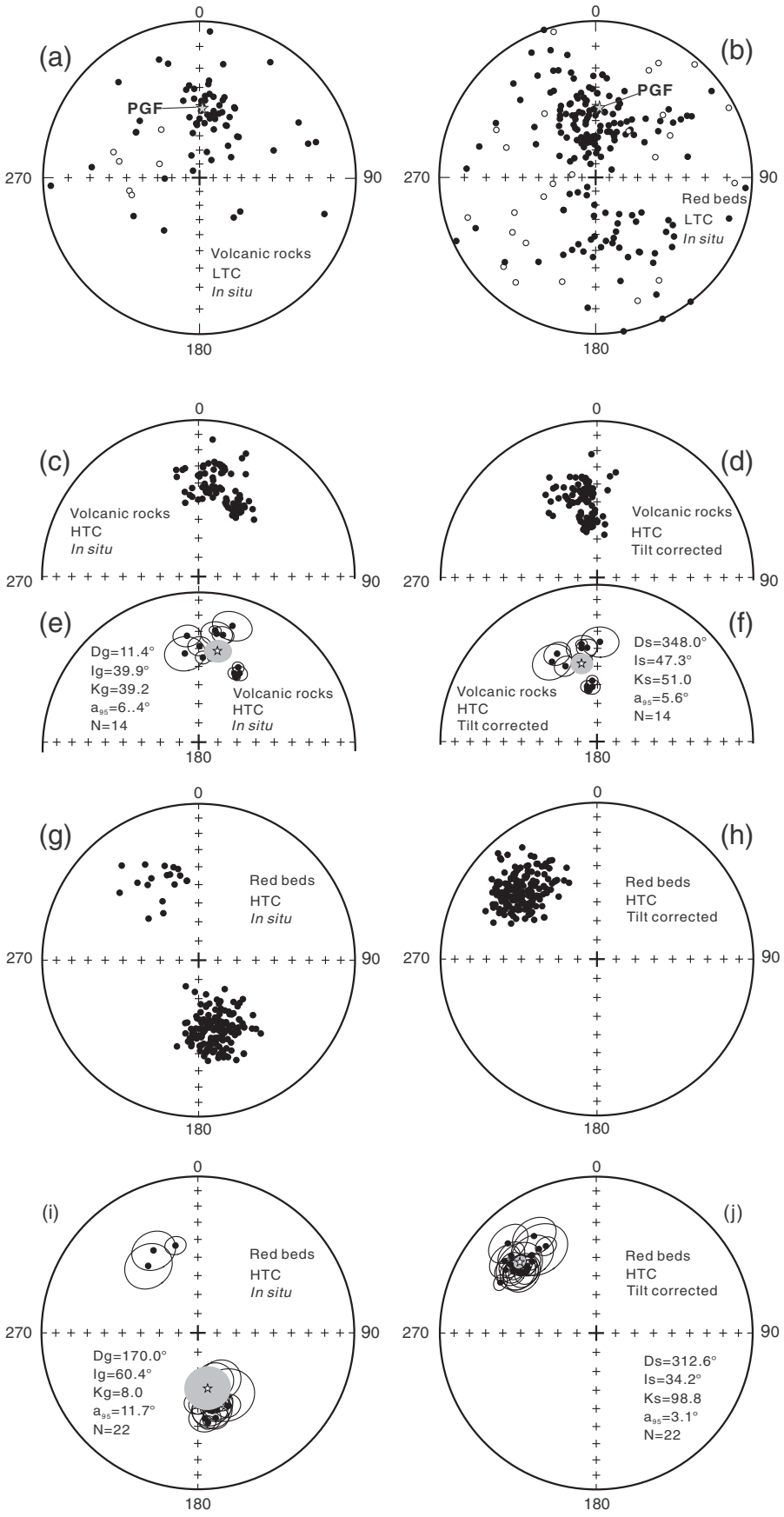


Fig. 8. Equal area projections of paleomagnetic directions. LTC: Low temperature component; HTC: High temperature component; PGF: direction of the present geomagnetic field. (a), (b), (c), (d), (g) and (h) are in sample level; (e), (f), (i) and (j) are in site level, five point stars indicate site-mean directions, ellipses represent the 95% confidence circle of a mean direction.

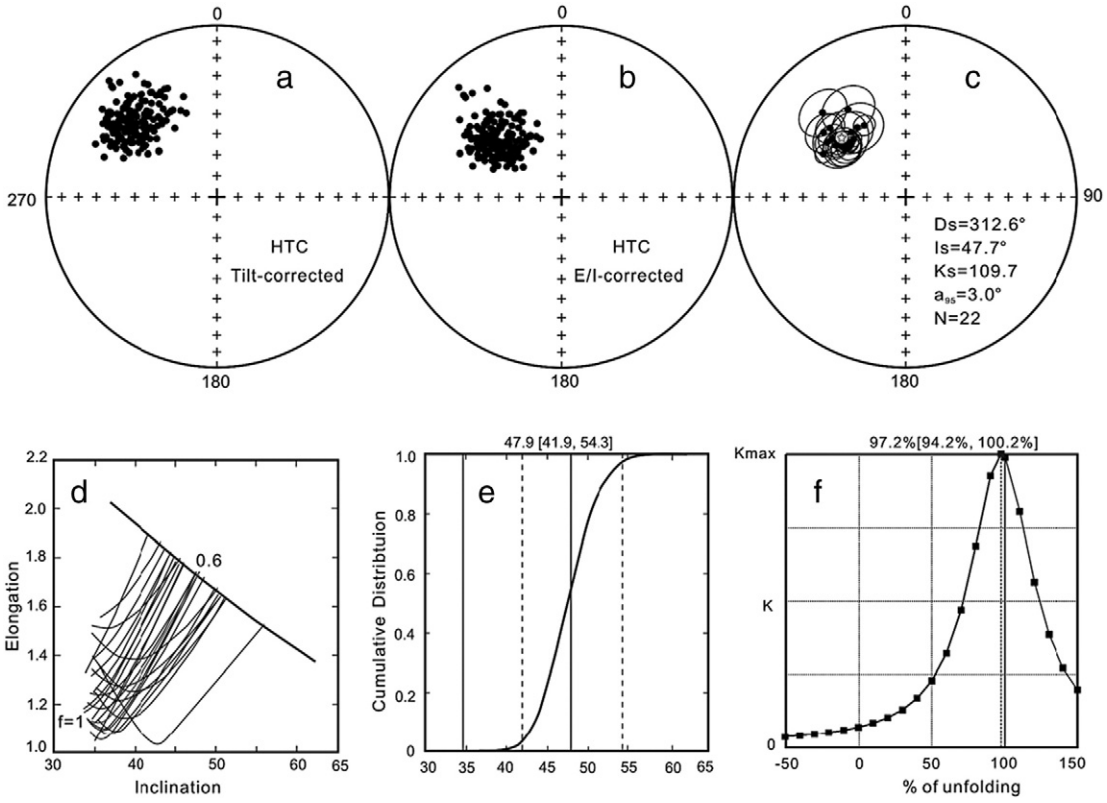


Fig. 9. (a) Equal area projections of the HTC direction from red beds in stratigraphic coordinates. (b) Results after the elongation/inclination (E/I) correction (Tauxe and Kent, 2004), plotted at sample level, $N = 174$. (c) After E/I correction, plotted at site level. (c) Results after E/I correction plotted at site level, the star indicating the site-mean direction with the circle of 95% confidence. (d) Plot of elongation *vs.* inclination as a function of flattening factor (f). (e) Cumulative distribution of the corrected inclinations. (f) Stepwise unfolding of the HTC directions after E/I correction.

terranes has happened since 100 Ma, along the Bangong–Nujiang suture. There is no significant paleolatitudinal difference observed between the Gerze red beds and upper Shexing Formation, suggesting that the convergence likely ceased by deposition of the Late Cretaceous red beds.

Our paleomagnetic analysis is well consistent with the convergence models reported recently (Li et al., 2010, 2013; Fan et al., 2014; C.S. Wang et al., 2014; Q. Wang et al., 2014; K.J. Zhang et al., 2014; Z.M. Zhang et al., 2014; Fan et al., 2015), which is largely based on geological evidence. Li et al. (2013) suggested that the oceanic lithosphere completely subducted between ~100–80 Ma and the significant thickening of the entire lithosphere happened during the Late Cretaceous caused by the Lhasa–Qiangtang collision. Moreover, Li et al. (2015a,b) considered that the Abushan volcanic rocks (~100 Ma) represent the magmatism caused by the Lhasa–Qiangtang collision and significant shortening (~650 km) that happened during the Late Cretaceous collision.

6.3. Vertical-axis rotation observed in the Tibetan Plateau

Paleomagnetic declinations may represent the cumulative amount of rotation since the remanence was acquired. The difference between an observed declination and expected ones can provide reliable kinetic evidence for the research of tectonic deformation. The expected direction of a specific locality could be calculated using the appropriate age reference paleomagnetic pole of the stable Asia (e.g., Torsvik et al., 2012; Cogné et al., 2013). We have tested that the difference between expected declinations calculated from the APWPs proposed by Torsvik et al. (2012) and Cogné et al. (2013) is not significant, and we adopt Cogné et al.'s (2013) APWP as reference to calculate the expected declination and paleolatitude (Table 3). Table 3 lists the observed directions of the Cretaceous rocks calculated from the reliable poles of the central Tibetan Plateau and reference poles of the East Asia. In the study of Huang et al. (1992), two directions were reported which indicated that the southern margin of the Qiangtang terrane had experienced a clockwise rotation of 47.1° ± 14.1° since the Berriasian–Barremian (145–125 Ma) and 28.5° ± 11.7° since the Aptian–Turonian (125–88.6 Ma) compared with expected directions from the East Asia, respectively (Table 3; Fig. 1a). The results of Otofuji et al. (1990) show the similar rotation model which suggests that the Markham area had undergone a clockwise rotation of 44.5° ± 11.1° since Early Cretaceous.

The paleomagnetic directions obtained from this study show more light on the tectonic rotational history of the Qiangtang terrane. As shown in Table 3, the observed paleomagnetic directions from the red beds demonstrate that the sampling area has rotated 55.0° ± 4.9° counterclockwise with respect to the East Asia reference pole after the deposition of the red beds (Fig. 10a), meanwhile the data from the ~110–100 Ma volcanic rocks suggest a 19.5° ± 7.6° counterclockwise rotation with respect to the coeval reference pole of East Asia (Fig. 10a). This large discrepancy may suggest that there may be considerable age difference between the red beds and the lavas, and ~35° clockwise rotation had happened in this time interval.

The rotation pattern of the Qiangtang terrane holds critical clues for the understanding of the regional tectonics. The clockwise rotation signature observed by this study is consistent in both amount and sense with those previously reported, suggesting that the clockwise rotation happened during the mid-Cretaceous but stopped during the Late Cretaceous. It may reflect the oblique collision of the Lhasa and Qiangtang terranes in Cretaceous, or right-lateral displacement of faults within the Bangong–Nongjiang suture zone (Dewey et al., 1988; Matte et al., 1996; Yin and Harrison, 2000; Wang et al., 2005; C.S. Wang et al., 2014; Q. Wang et al., 2014). But the following large scale counterclockwise rotation revealed from our results is likely due to the Indo–Asia collision, because the moving velocity of the northeast corner of the India plate was always greater than the moving velocity of the northwest corner (Molnar and

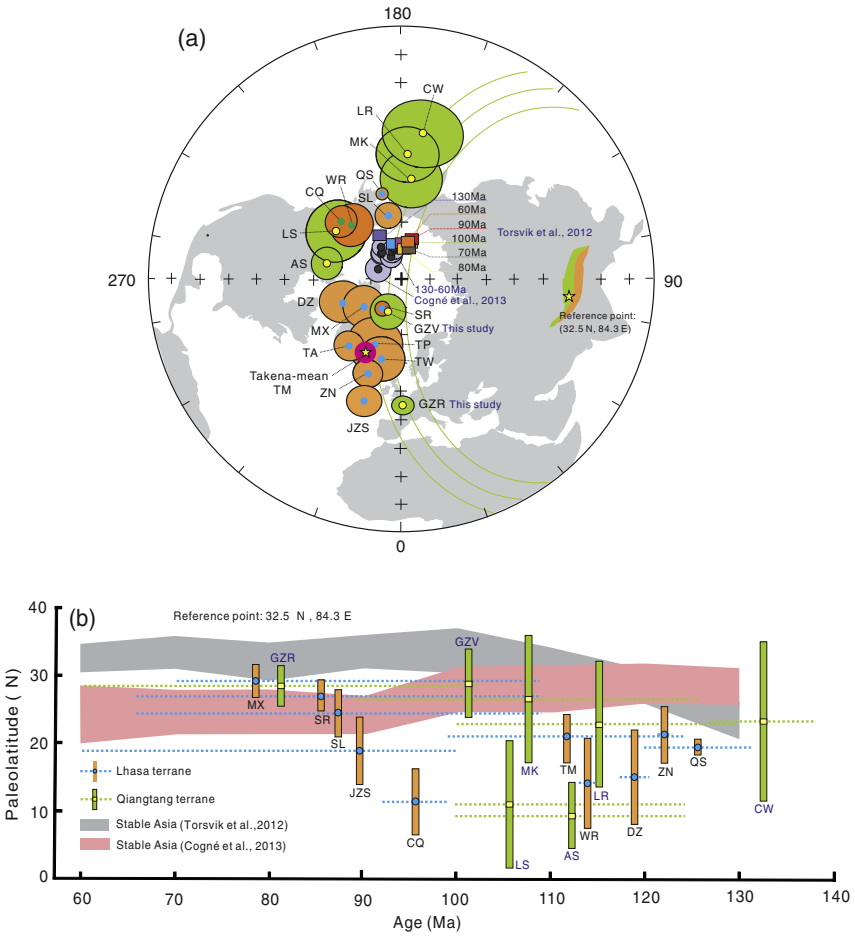


Fig. 10. (a) Equal-area projections of the high quality Cretaceous paleomagnetic poles (listed in Table 2) from the Qiangtang and Lhasa terranes with the reference locations (Gerze); arcs of the small circles are calculated from the ~100 Ma Gerze lava pole and its uncertainty at 95% confidence; yellow/green dots with shaded 95% confidence circle are Cretaceous poles from the Qiangtang/Lhasa terranes; colorized rectangles are the poles of Asia during 130–60 Ma following Torsvik et al. (2012); gray dots with blue 95% confidence circle are the poles of Asia during 130–60 Ma following Cogné et al. (2013). (b) Plots of Cretaceous paleolatitudes of Asia, Qiangtang and Lhasa terranes at Gerze (reference point: 32.5°N, 84.3°E) are calculated from the poles plotted in (a); the expected paleolatitudes of the stable Asia are calculated from APWPs (Torsvik et al., 2012; Cogné et al., 2013) while the paleolatitudes of the Qiangtang and Lhasa terranes are calculated from reliable poles (Fig. 10a); vertical bars and dot aclinic line display paleolatitude and age uncertainties, respectively.

Tapponnier, 1975). This discrepancy of moving velocity led to a large-scale counterclockwise rotation in many locations in the Tibetan Plateau when the India plate collided with the Asian plate (Fig. 1a).

In the Lhasa terrane, the Cretaceous paleomagnetic declinations show a remarkably consistent rotated trend (Fig. 1a; Table 3). Most studies suggest that the area of the central Lhasa had experienced a different amount (20°–40°) of anticlockwise rotation since the Early Cretaceous (Pozzi et al., 1982; Westphal et al., 1983; Achache et al., 1984; Tan et al., 2010; Chen et al., 2012; Sun et al., 2012), except for four results which indicate a slight clockwise rotation of the middle-west part of the Lhasa terrane calculated from poles SL, CQ, WR and QS (Fig. 1a; Table 3) (Sun et al., 2008; Tan et al., 2010; Tang et al., 2013; Ma et al., 2014). As one of the most active tectonic region, the western Tibetan was not only controlled by the collision but was also affected by some Cenozoic sinistral strike–slip faults. Therefore, the discrepancy between Cretaceous paleomagnetic declinations from the Lhasa terrane could be explained by these kinds of faults which caused some local rotation around different sampling areas (Yin and Harrison, 2000; Kapp et al., 2007; Ratschbacher et al., 2011; Yin and Taylor, 2011).

7. Conclusive remarks

1. We dated a sequence of volcanic rocks at ~110–100 Ma, using zircon U–Pb method, and carried out a paleomagnetic investigation for these rocks and the overlying red beds that are likely of Late Cretaceous in age.
2. Paleomagnetic results from both volcanic rocks and red beds passed a fold test, respectively; paleosecular variation has been averaged out for the directions observed from the volcanic rocks and inclination shallowing test and E/I correction have been performed for the red beds.
3. The new paleomagnetic results indicate that the Qiangtang terrane was a tectonic coherent part of the Asian continent by ~110 Ma.
4. At least ~550 km N–S convergence between the Qiangtang and Lhasa terranes happened after ~100 Ma.
5. The Gerze area had experienced a significant counterclockwise rotation after ~100 Ma, which is most likely caused by the India–Asia collision.

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.gr.2015.07.004.

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Table 3
Cretaceous paleomagnetic directions and the amount of rotation with expected directions calculated from the stable Eurasia.

Mnemonic	Site Location (°N/°E)	Age (Ma)	Observed Direction			Reference pole (Eurasia)			Dec-exp.	Rotation	Lat-exp.	Displacement	Reference
			Dec. (°)	Inc. (°)	α_{95} (°)	Lat. (°N)	Lon. (°E)	A_{95} (°)	(°)	R ± ΔR(°)	(°)	D ± ΔD(°)	
Qiangtang													
GZR	32.5/84.3	60 –103.8	312.6	47.7	3.0	80.5	219.5	2.9	7.4	−54.8 ± 4.9	25.5	−3.2 ± 3.5	This study
GZV	32.5/84.3	103.8 ± 0.46	348.0	47.3	5.6	82.2	205.4	3.5	7.6	−19.6 ± 7.6	28.2	−1.1 ± 5.4	This study
MK	29.7/98.7	88.6–125.0	39.0	52.5	8.5	80.3	204.0	1.9	10.5	28.5 ± 11.7	26.7	−6.3 ± 7.8	Huang et al. (1992)
CW	29.7/98.4	125.0–145.0	58.1	51.0	10.9	80.3	193.2	1.5	11.0	47.1 ± 14.1	28.4	−3.2 ± 1.5	Huang et al. (1992)
LR	29.7/98.6	99.6–130	48.2	49.0	8.8	80.9	197.4	1.5	10.2	44.5 ± 11.1	27.9	−2.0 ± 7.7	Otofuji et al. (1990)
AS	35.0/79.7	99.6–125.0	1.3	21.8	5.0	80.9	197.4	1.5	9.3	−8.0 ± 5.5	30.4	19.1 ± 4.3	Chen et al. (1993)
LS	34.5/80.4	99.6–125.0	12.4	22.2	9.7	80.9	197.4	1.5	9.4	3.0 ± 10.3	30.0	18.5 ± 8.0	Chen et al. (1993)
Lhasa													
JZS	31.2/84.7	60.0–100.0	317.6	31.2	5.3	80.5	219.5	2.9	7.4	−49.8 ± 6.4	24.3	7.5 ± 4.8	Yang et al. (2015)
MX	29.9/90.7	72.4–110.0	343.5	43.1	2.6	80.1	211.4	2.0	9.4	−25.9 ± 3.7	24.5	−0.6 ± 2.7	Sun et al. (2012), Lippert et al. (2014)
SR	29.9/91.1	65.0–110.0	350.2	41.5	2.5	80.5	219.5	2.9	8.1	−17.9 ± 4.1	23.4	−0.5 ± 3.0	Tan et al. (2010)
SL	29.9/91.2	65.0–110.0	22.6	41.5	4.4	80.5	219.5	2.9	8.1	14.4 ± 5.6	23.5	−0.4 ± 4.1	Tan et al. (2010)
TP	29.2/90.0	99.6–125.0	338.0	35.9	10.0	80.9	197.4	1.5	9.7	−31.7 ± 9.5	26.1	6.2 ± 7.1	Pozzi et al. (1982)
TW	29.9/91.0	99.6–125.0	332.7	37.9	8.0	80.9	197.4	1.5	9.8	−37.1 ± 8.0	27.0	5.7 ± 6.0	Westphal et al. (1983)
TNA	31.0/92.0	99.6–125.0	339.0	24.3	8.9	80.9	197.4	1.5	10.0	−31.6 ± 6.9	28.2	14.7 ± 5.3	Achache et al. (1984)
TSA	30.0/91.0	99.6–125.0	338.8	23.2	8.3	80.9	197.4	1.5	9.8	−15.5 ± 8.2	27.1	15.1 ± 6.4	Achache et al. (1984)
WR	30.5/90.1	114.0 ± 1.1	18.4	26.6	8.6	80.8	199.5	3.5	9.8	8.6 ± 8.1	27.1	13.1 ± 6.2	Sun et al. (2008)
DZ	31.1/84.4	117.0–121.0	350.6	25.9	7.4	80.6	192.9	2.9	10.1	−19.5 ± 8.3	27.7	14.1 ± 6.4	Yang et al. (2015)
ZN	31.3/85.0	123.1 ± 0.9	326.9	35.8	4.5	80.3	193.2	1.5	10.4	−43.5 ± 5.2	27.8	8.0 ± 3.9	Chen et al. (2012)
QS	32.3/82.6	120–132	28.4	34.8	2.0	80.3	193.2	1.5	10.3	18.0 ± 2.8	28.5	9.3 ± 2.1	Ma et al. (2014)

Notes: Mnemonic: pole name (details see Table 2); Site location: paleomagnetic sampling locality; Age: see Table 2 and compare the geological stages with the Geological Time Scale (Gradstein et al., 2012); Dec/Inc: declination/inclination calculated from the responding pole; α_{95}: radius of the 95% confidence circle of a direction; Reference pole: the coeval Eurasia reference pole chosen (or averaged) for the corresponding age (or age interval), based on the data of Cogné et al. (2013); Lat/Lon: latitude/longitude of the reference pole; A_{95}: radius of the 95% confidence circle of a pole; Dec-exp./Lat-exp.: declination/latitude expected at the sampling site from the coeval Eurasia reference pole; Rotation (R ± ΔR): the vertical axis rotation calculated with respect to the expected declination with 95% confidence limit (+/−: clockwise/counterclockwise); Displacement (D ± ΔD): the north–south displacement calculated with respect to the expected latitude with 95% confidence limit.

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